UNIVERSITI TEKNOLOGI PETRONAS

TEB2043: DATA SCIENCE

JANUARY 2026

GROUP 12 DASHBOARD

PROJECT TITLE: ANALYZING LIFESTYLE AND DIETARY FACTORS INFLUENCING OBESITY LEVELS

INSTRUCTOR: MUHAMMAD HAMZA AZAM

BY

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Dashboard Setup

1. Make sure these files exist in the same folder
    - obesity_dashboard.html
    - dashboard_api.py
    - train.csv
    - test.csv
2. Start the server:

    python "c:\...\ dashboard_api.py"
3. Open the dashboard in your browser through Visual Studio Code by setting up a live server for obesity_dashboard.html
4. All files can be found in this GitHub repository:

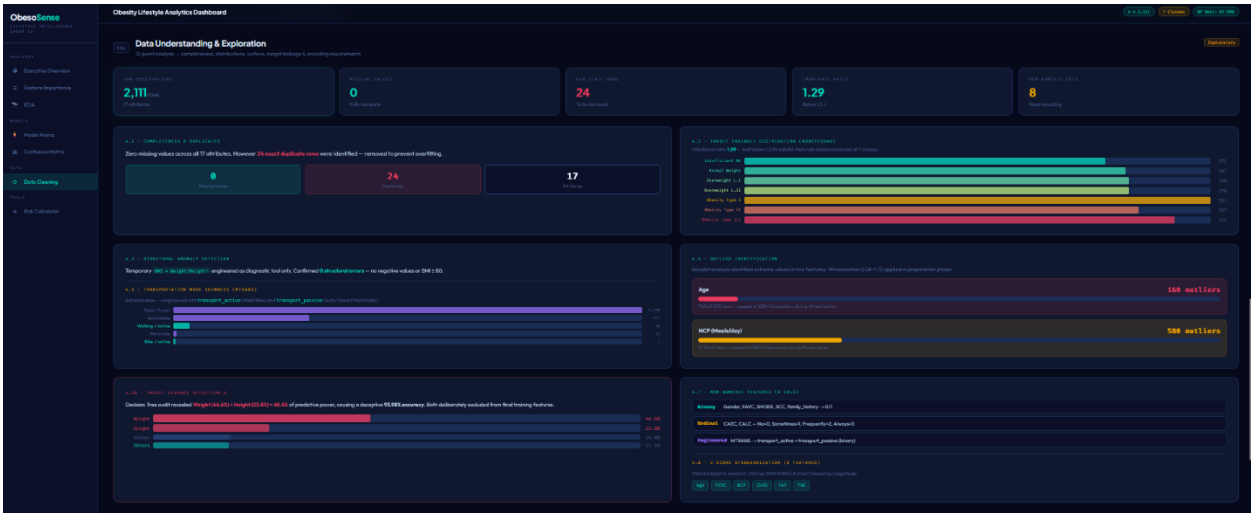
    https://github.com/AmmirrulAmran/Data-Science-Project-on-Obesity-Prediction-Based-on-Lifestyle

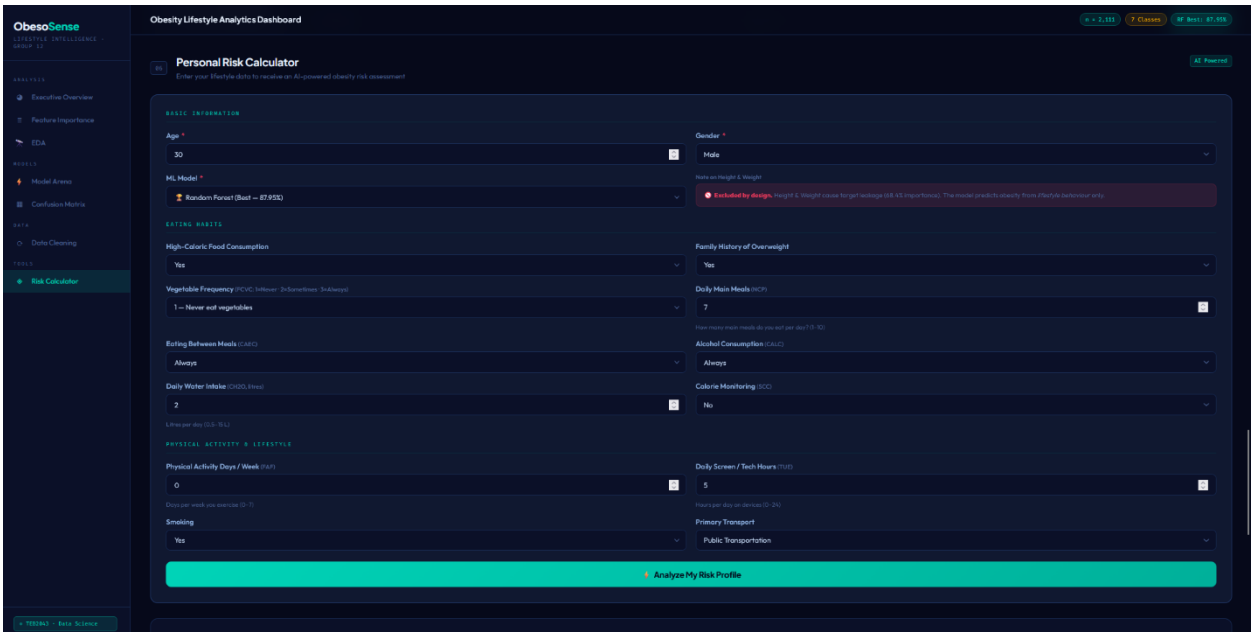
Dashboard features

- **Executive Overview**
  - Donut chart + legend showing the **7 obesity class distribution** (counts and percentages).
  - Quick headline stats (sample size, best accuracy, feature count).
- **Feature Importance**
  - Ranks the most influential lifestyle features (shown as a Random Forest style importance list).
  - Visual comparison chart for importance (used for interpretation, not training live).
- **EDA (Exploratory Data Analysis)**
  - "Audit cards" summarizing dataset health: **missing values, duplicates, imbalance ratio, non-numeric columns**.
  - Target distribution breakdown by class.
  - Highlights structural checks (e.g., BMI as a diagnostic concept) and outlier notes.
- **Model Arena**
  - Lets you compare models (Decision Tree, Random Forest, KNN, SVM, Logistic Regression) using **precomputed performance metrics** (test vs cross-validation accuracy).
  - Purpose: quick "which model is best" overview.
- **Confusion Matrix Heatmap**
  - Interactive confusion matrix section (with model selector buttons) that visualizes where predictions go wrong (diagonal = correct, off-diagonal = errors).
  - Class label list + "key observations" text.
- **Data Cleaning Pipeline (interactive)**
  - A step-by-step view that shows how the dataset is transformed:
    - **Duplicate removal**
    - **Structural error checks / invalid row filtering**
    - **Encoding non-numeric features** (binary + ordinal)
    - **Outlier treatment (winsorization)** notes

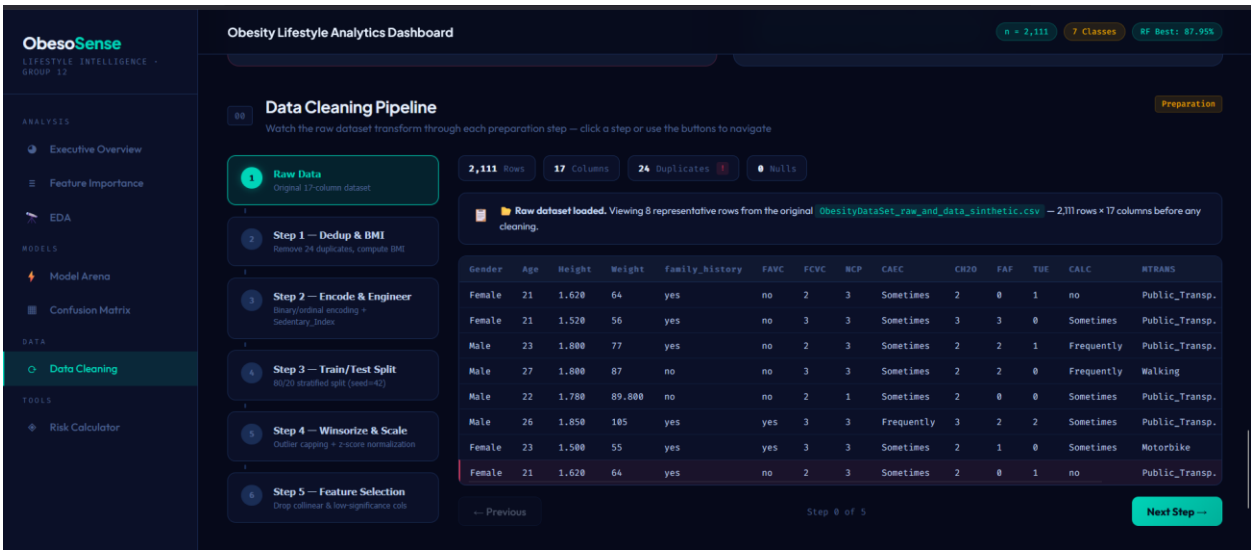
        - **Scaling/standardization** notes
        - **Feature selection** (correlation drop + leakage prevention like dropping height/weight)
    - Displays a changing table + step progress UI so users can “watch” the data being cleaned.
- **Personal Risk Calculator**
    - Form for lifestyle inputs (age, habits, water intake, activity, screen time, etc.).
    - Predicts one of the **7 obesity classes** and shows:
        - Predicted class + risk level (low → critical)
        - Probability bars for all classes
        - Auto-generated explanation + “key risk factors”
    - With the Python server running (dashboard_api.py), the dropdown truly switches between **all 5 ML models** for prediction.

Dashboard Preview


ObesoSense
Obesity Lifestyle Analytics Dashboard
Data Understanding & Exploration
2,111
0
24
1.29
8
17


ObesoSense
Obesity Lifestyle Analytics Dashboard
Personal Risk Calculator
Executive Overview
Feature Importance
EDA
Model Arena
Confusion Matrix
Data Cleaning
Risk Calculator
BASIC INFORMATION
Age
30
Gender
Male
ML Model
Random Forest (Best — 87.95%)
EATING HABITS
High-Caloric Food Consumption
Yes
Family History of Overweight
Yes
Vegetable Frequency
1 — Never eat vegetables
Daily Main Meals
7
Eating Between Meals
Always
Alcohol Consumption
Always
Daily Water Intake
2
Calorie Monitoring
No
PHYSICAL ACTIVITY & LIFESTYLE
Physical Activity Days / Week
0
Daily Screen / Tech Hours
5
Smoking
Yes
Primary Transport
Public Transportation
Analyze My Risk Profile

Executive Overview
Feature Importance
EDA
Model Arena
Confusion Matrix
Data Cleaning
Risk Calculator
Overweight Level II
The model identifies Overweight Level II with 21% confidence. Your sedentary index (5.00), snacking habit (Always), and screen time (5h/day) are primary drivers. Target 3–4 active days/week and reduce high-caloric food intake.
Low activity: only 0 day(s)/week exercise
High sedentary index: 5 (screen time vs activity)
Insufficient Weight 13.5%
Normal Weight 16.5%
Overweight Level I 20.0%
Overweight Level II 20.5%
Obesity Type I 17.5%
Obesity Type II 12.0%
Obesity Type III 0.0%


ObesoSense
LIFESTYLE INTELLIGENCE · GROUP 12
Obesity Lifestyle Analytics Dashboard
n = 2,111
7 Classes
RF Best: 87.95%
ANALYSIS
Executive Overview
Feature Importance
EDA
MODELS
Model Arena
Confusion Matrix
DATA
Data Cleaning
TOOLS
Risk Calculator
00
Data Cleaning Pipeline
Preparation
Watch the raw dataset transform through each preparation step — click a step or use the buttons to navigate
1 Raw Data
Original 17-column dataset
2 Step 1 — Dedup & BMI
Remove 24 duplicates, compute BMI
3 Step 2 — Encode & Engineer
Binary/ordinal encoding + Sedentary_Index
4 Step 3 — Train/Test Split
80/20 stratified split (seed=42)
5 Step 4 — Winsorize & Scale
Outlier capping + z-score normalization
6 Step 5 — Feature Selection
Drop collinear & low-significance cols
2,111 Rows
17 Columns
24 Duplicates !
0 Nulls
Raw dataset loaded. Viewing 8 representative rows from the original ObesityDataSet_raw_and_data_sinthetic.csv — 2,111 rows × 17 columns before any cleaning.
Gender Age Height Weight family_history FAVC FCVC NCP CAEC CH2O FAF TUE CALC MTRANS
Female 21 1.620 64 yes no 2 3 Sometimes 2 0 1 no Public_Transp.
Female 21 1.520 56 yes no 3 3 Sometimes 3 3 0 Sometimes Public_Transp.
Male 23 1.800 77 yes no 2 3 Sometimes 2 2 1 Frequently Public_Transp.
Male 27 1.800 87 no no 3 3 Sometimes 2 2 0 Frequently Walking
Male 22 1.780 89.800 no no 2 1 Sometimes 2 0 0 Sometimes Public_Transp.
Male 26 1.850 105 yes yes 3 3 Frequently 3 2 2 Sometimes Public_Transp.
Female 23 1.500 55 yes yes 3 3 Sometimes 2 1 0 Sometimes Motorbike
Female 21 1.620 64 yes no 2 3 Sometimes 2 0 1 no Public_Transp.
← Previous
Step 0 of 5
Next Step →